10.1 Attributes of Industrial Revolution

- Use of new basic materials: iron and steel
- Use of new energy sources: coal, electricity, petroleum
- Invention of new machines such as the spinning jenny and the power loom that increased the production with a minimum expending of human energy
- Emergence of a new organization known as the factory system, which entailed increased division of labour and specialisation of work
- Development in transportation and communication
- Increasing application of science to industry
- The use of new technology

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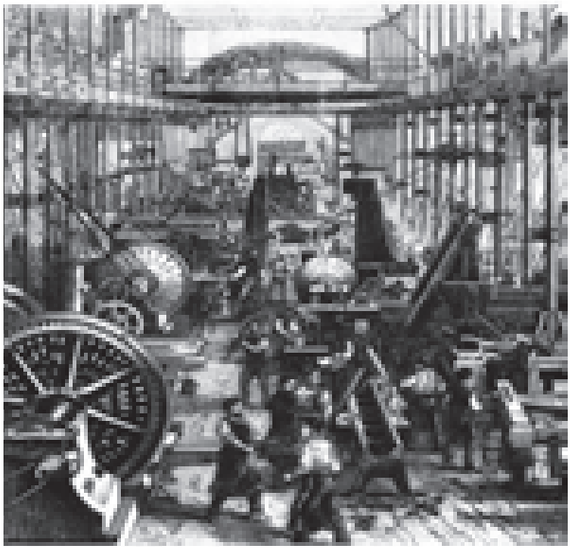
Beginnings

The Industrial Revolution began in England first because, it had certain external factors. They were:

- England had abundant resources and possessed colonies, with "India being the brightest jewel in the British Crown"
- Access to coal, iron and raw cotton from the colonies
- England possessed the required infrastructure for textiles, developed by immigrant artisans from the Netherlands
- England had a developed banking system, a growing entrepreneurial class, and potential investors
- Encouragement of the Royal Society of England for scientific discoveries and inventions
- Political stability of England to bestow its full attention to industrial growth

Invention of Steam Power

In the 18th Century, British mine-owners were faced with the problem of water seeping into the mines. Water had to be removed to extract coal. So they employed additional labourers to pump the water out. Employing human labour cost a lot of money. It was at that juncture the British engineer, Thomas Newcomen invented a contrive to pump the water out of mines. But the mechanism he developed consumed too much fuel. James Watt, a Scottish engineer, converted a stationary steam engine to a rotary engine which consumed less fuel.



A scene in an English factory

Development in Textiles

Before the Industrial Revolution, the spinning and weaving of cloth were undertaken for domestic and local consumption. It was done at home or in a small hired place. The production also took place on a cottage scale. The manually operated spinning wheel required four to eight spinners to supply yarn to one handloom weaver. In 1733 John Kay invented the 'Flying Shuttle' which, when operated by hand, increased the speed of the weaving of cloth. In 1764 James Hargreaves invented 'the spinning jenny'. This machine spun eight threads at one and the same time. Two years later Richard Arkwright invented

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